Radial Velocity Characterization of the near-resonance TOI-233 System

Results Summary

Camilo Ospinal
Universidad de Antioquia

Supervised by Emma Turtelboom and Ryan Cloutier
McMaster University

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Summary

We present the results of the radial velocity (RV) analysis of the TOI-233 system for its two near-resonance planet candidates, TOI-233.02 ($P_2 \approx 7.20$ days) and TOI-233.01 ($P_1 \approx 11.67$ days). Using the Line-by-Line (LBL) method, we extracted RVs from 25 MAROON-X spectra, with RMS ≈ 2.892 m/s and $\Delta\mathrm{RV}_{\mathrm{mean}} \approx 1.35$ m/s. The periodogram of the MAROON-X RVs recovers not only the orbital periods but also the stellar rotation period, $P_{\mathrm{rot}} \approx 43$ days. We then used the RadVel package to find the best fit for the combined MAROON-X and ESPRESSO data, applying a quasi-periodic Gaussian Process. The favored model is the one with free parameters K_1, K_2, GP, and γ, giving final planet masses of $M_1 \sin i = 2.82 \pm 0.72$ $\mathrm{M}_\oplus$ and $M_2 \sin i = 2.81 \pm 0.59$ $\mathrm{M}_\oplus$. These masses, along with their possible radii and calculated bulk densities, suggest that these planets could be part of the group of puffy planets in near mean resonance.

Data and Model

We have 25 MAROON-X observations, reduced by the Gemini North team without telluric correction. Alongside these spectra, we also have the radial velocity extraction results from the SERVAL pipeline, where the blue and red channels were reduced separately. SERVAL also provides activity indicators such as the differential linewidth of observation (dLW), the chromatic index of observation (CRX), and the Hα activity index, among others. The mean RV uncertainties from SERVAL for the blue and red channels are $\Delta\mathrm{RV}_{\mathrm{B,mean}} = 1.275$ m/s and $\Delta\mathrm{RV}_{\mathrm{R,mean}} = 1.226$ m/s, respectively, and the RMS values are $\mathrm{RMS}_{\mathrm{B}} = 2.661$ m/s and $\mathrm{RMS}_{\mathrm{R}} = 2.889$ m/s.

With the aim of mitigating possible stellar-activity effects on the individual radial velocities of TOI-233's spectral lines, and given its nature as a red dwarf ($M_* = 0.37$ $\mathrm{M}_\odot$), we used the Line-by-Line method (e.g., Bouchy et al. 2001; Dumusque 2018; Cretignier et al. 2020) via the LBL package[1] (Artigau et al. 2022), which is optimized to obtain precision radial velocities (pRV) for late-type stars. Assuming a stellar temperature of 3568 K (Stassun et al. 2019), we extracted the RVs of each channel separately, letting LBL compute its own self-templates from the 25 observations and the telluric correction. We compared these results against a new LBL extraction using a friend-template built from 70 spectra of BD+443567, an M2V star with $T \approx 3600$ K, provided by Alexandrine L'Heureux

[1] https://lbl.exoplanets.ca/

Table 1: Parameters setted in the Keplerian model, instrumental terms, and Gaussian Process (GP) using RadVel.

Keplerian Model					
Parameter	Description	Initial value	Prior	Bounds / scale	Unit
`time_base`	Reference epoch	60954.388	Fixed	–	BJD
P_1	Orbital period of TOI-233.01	11.6700285	Gaussian	$\mu = P_1,\ \sigma = 0.0001$	d
$t_{c,1}^{\rm orig}$	Original conjunction time of TOI-233.01	60956.008	Fixed	–	BJD
$t_{c,1}$	Relative conjunction time (RadVel)	1.620	Gaussian	$\mu = 1.620,\ \sigma = 1.0$	d
$\sqrt{e_1}\cos\omega_1$	Eccentricity component	0	HardBounds	$[-1, 1]$	–
$\sqrt{e_1}\sin\omega_1$	Eccentricity component	0	HardBounds	$[-1, 1]$	–
$\ln K_1$	Log RV semi-amplitude	$\ln(3.0)$	HardBounds	$[\ln(0.5), \ln(20)]$	–
P_2	Orbital period of TOI-233.02	7.2011383	Gaussian	$\mu = P_2,\ \sigma = 0.0001$	d
$t_{c,2}^{\rm orig}$	Original conjunction time of TOI-233.02	60951.881	Fixed	–	BJD
$t_{c,2}$	Relative conjunction time (RadVel)	-2.507	Gaussian	$\mu = -2.507,\ \sigma = 1.0$	d
$\sqrt{e_2}\cos\omega_2$	Eccentricity component	0	HardBounds	$[-1, 1]$	–
$\sqrt{e_2}\sin\omega_2$	Eccentricity component	0	HardBounds	$[-1, 1]$	–
$\ln K_2$	Log RV semi-amplitude	$\ln(2.0)$	HardBounds	$[\ln(1), \ln(10)]$	–
e	Eccentricity prior	–	EccentricityPrior	0.159	–
Instrumental Terms					
Parameter	Description	Initial value	Prior	Bounds / scale	Unit
$\gamma_{\rm MX,B}$	RV offset MAROON-X Blue	0	HardBounds	$[-50, 50]$	m s^{-1}
$\gamma_{\rm MX,R}$	RV offset MAROON-X Red	0	HardBounds	$[-50, 50]$	m s^{-1}
$\gamma_{\rm ESP}$	RV offset ESPRESSO	0	HardBounds	$[-50, 50]$	m s^{-1}
$\sigma_{\rm jit,MX,B}$	Jitter MAROON-X Blue	2	HardBounds	$[0, 10]$	m s^{-1}
$\sigma_{\rm jit,MX,R}$	Jitter MAROON-X Red	2	HardBounds	$[0, 10]$	m s^{-1}
$\sigma_{\rm jit,ESP}$	Jitter ESPRESSO	2	HardBounds	$[0, 10]$	m s^{-1}
$\dot{\gamma}$	Linear trend	0	Fixed	0	m s^{-1} d^{-1}
$\ddot{\gamma}$	Curvature	0	Fixed	0	m s^{-1} d^{-2}
Gaussian Process: quasi-periodic kernel					
Parameter	Description	Initial value	Prior	Bounds / scale	Unit
$A_{\rm GP}$ or η_1	GP amplitude	26.0	Jeffreys	$[0.01, 100]$	m s^{-1}
$P_{\rm GP}$ or η_2	GP characteristic period	43.5	HardBounds	$[0.1, 100]$	d
$\ell_{\rm exp}$ or η_3	Exponential timescale	$43.5\sqrt{2}$	HardBounds	$[0.1, 100]$	d
$\ell_{\rm per}$ or η_4	Periodic coherence scale	$\sqrt{1/(2 \times 3.32)}$	HardBounds	$[0.01, 10]$	–

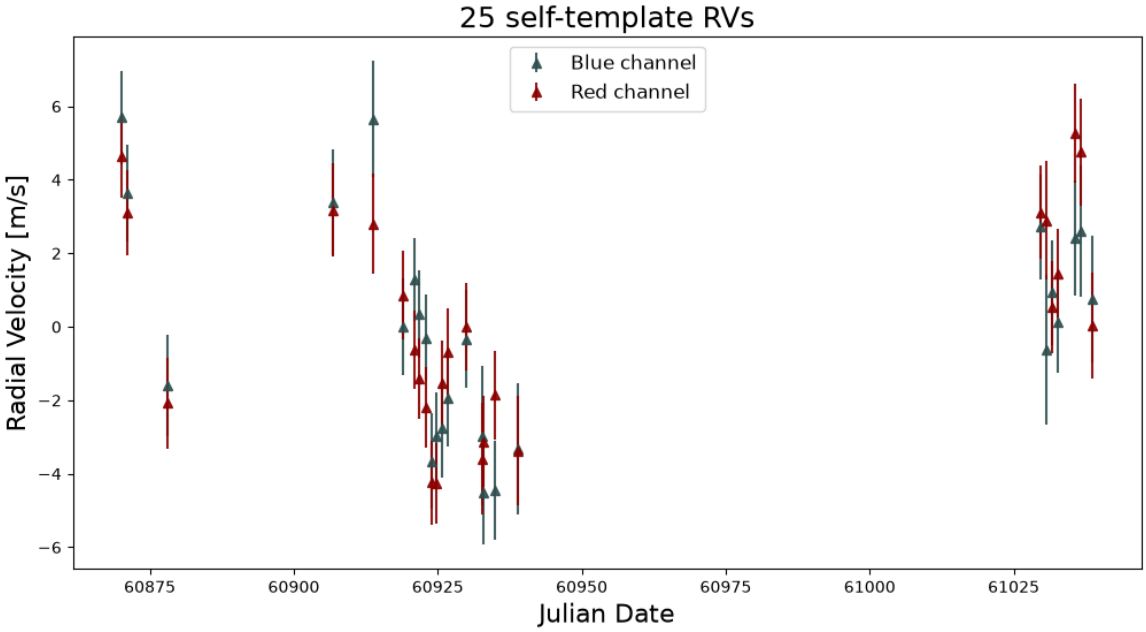


Figure 1: Caption

(also using LBL). We then computed the Lomb-Scargle periodograms of the RVs and of several activity indicators.

In addition to the MAROON-X spectra, we also have data from ESPRESSO and HARPS. For each of these instruments we computed the mean uncertainty, the RMS, and the information content, σ_K, following Equation 1 (Cloutier et al. 2018),

$$\sigma_K = \sigma_{\rm RV}\sqrt{\frac{2}{N_{\rm RV}}}, \tag{1}$$

where $\sigma_{\rm RV}$ was taken to be the RMS and $N_{\rm RV}$ is the number of RV data points.

To compute the system's radial velocity model, we used the MAROON-X data together with 15 ESPRESSO radial velocity measurements, for a total of 65 data points, treating the MAROON-X channels as independent instruments within the RadVel package[2]. We assumed a 2-planet Keplerian model with a stellar mass of 0.37 $M_\odot$ and periods of $P_2 = 7.20$ days and $P_1 = 11.67$ days, with Gaussian priors. We assigned each instrument its own zero-point, with an initial value of $\gamma_i = 0$ m/s and HardBounds priors, and, similarly, individual jitter terms with an initial value of 2.0 m/s. MAROON-X is known to have an etalon drift of $\sim$ 2–3 cm/s/day; however, since RadVel assigns the corresponding $\dot{\gamma}$ parameter to the instrument set as a whole rather than per instrument, we fixed it to zero with no variation. To ensure low eccentricities consistent with the system's near-resonance dynamics, we computed the crossing-eccentricity limit that sets the geometric criterion at which the orbits of two planets cross (Lammers et al. 2024), in our notation:

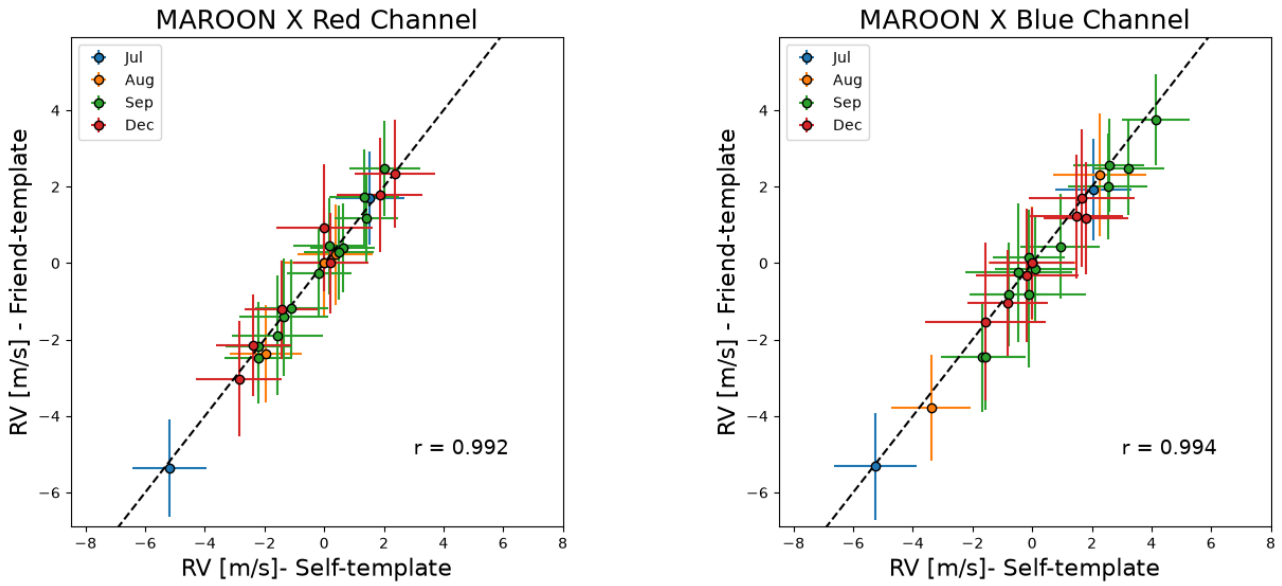
$$e_{\rm cross} = \frac{(P_1/P_2)^{2/3} - 1}{(P_1/P_2)^{2/3} + 1}. \tag{2}$$

[2]https://radvel.readthedocs.io/en/latest/index.html

Accordingly, we set the EccentricityPrior with $e_{\rm cross} = 0.159$ as the upper bound for both planets. Finally, we applied a quasi-periodic Gaussian Process to remove the effects of stellar activity. Table 1 summarizes the Keplerian, instrumental, and Gaussian Process parameters and their priors as configured in RadVel.

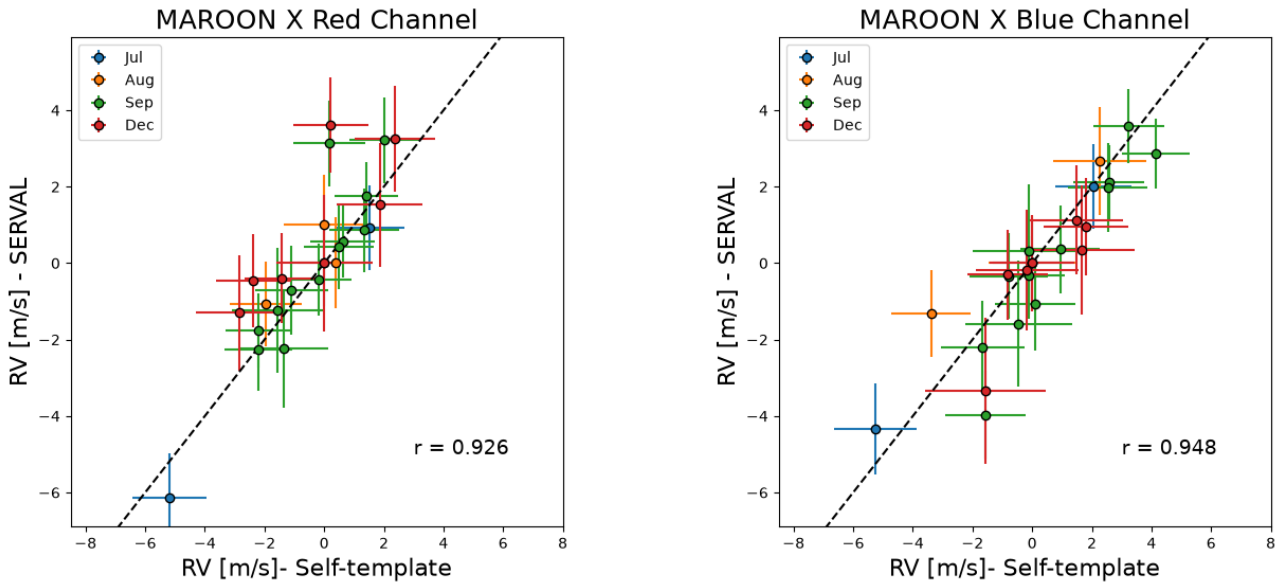
Finally, assuming radii of $R_1 = 2.10 \pm 0.10$ R$_\oplus$ and $R_2 = 1.65 \pm 0.13$ R$_\oplus$ —previously derived from TESS transit curves by Emma Turtelboom—we place these planets in the context of confirmed exoplanets with measured periods and masses less than Jupiter's mass, both with and without resonance. Using the gas dwarf model by Tang et al. (2025) and the steam world model by Aguichine et al. (2025), available in the Python package mr-plotter (Castro-González et al. 2023), we make an initial estimate of their internal composition. Using a sample of 508 confirmed exoplanets with measured periods and masses smaller than Jupiter's, of which 211 are in near-resonance, we selected those with a precision of 100 in their radius and mass. We configured six models for water-rich planets (steam worlds) orbiting a 10-Gyr-old M dwarf. For the first four, we defined a Water Mass Fraction (WMF) of 10%, 30%, 70%, and 90%, with an equilibrium temperature of $T_{\rm eq} = 400$ K. For the last two, we defined WMFs of 20% and 80%, with equilibrium temperatures of 424 and 361 K, as reported for TOI-233.02 and TOI-233.01, respectively, in the NASA Exoplanet Archive. For the H_2-He-rich planet model, we configured three models with solar metallicity in the planet's gaseous envelope, an age of 10 Gyr, $T_{\rm eq} = 361$ K, and envelope mass fractions of 0.6%, 0.8%, and 0.9%, respectively, and we discarded those model planets that had boiled-off.

Results



(a) (b)

Figure 2: Linear correlation diagram between the RVs extracted using the self-template and the friend-template for the MAROON-X red (left) and blue (right) channels. The dashed line represents the unit-slope line. The r value is the Pearson correlation coefficient.



(a) **(b)**

Figure 3: Same as Figure 2, comparing the SERVAL results with the self-template results.

Radial velocities Extraction

The time series of the relative RVs extracted from MAROON-X using the self-template, after subtracting the median, is shown in Figure 1. The mean uncertainties and RMS values for both channels with the self-template, friend-template, and SERVAL are given in Table 2. Overall, LBL does not reduce these values relative to SERVAL, and the difference between the self-template and the friend-template is not significant. Consistently, the RMS and uncertainties of the blue-channel RVs extracted with LBL are higher, which is most evident for $\mathrm{RMS_B}$.

Method	$\Delta\mathrm{RV_{R,mean}}$	$\Delta\mathrm{RV_{B,mean}}$	$\mathrm{RMS_R}$	$\mathrm{RMS_B}$
Self-template	1.257	1.433	2.891	2.983
Friend-template	1.324	1.480	2.848	2.911
SERVAL	1.226	1.275	2.889	2.661

Table 2: Mean uncertainties and RMS for the RVs of the two MAROON-X channels resulting from LBL with a self-template and a friend-template, and from SERVAL.

Figures 2 and 3 show a comparison between the self-template and friend-template RVs, and between SERVAL and the self-template RVs, respectively. In the first case we observe good agreement between both templates relative to the unit-slope line. In the second case we see larger scatter, especially in the data from December. For each comparison, the Pearson correlation coefficients are above 0.9, and they are also highest when correlating the two LBL extractions.

Given this similarity between the two templates, we continued using the self-template results for subsequent calculations. These radial velocities also have an information content of $\sigma_K = 0.82$ m/s for both channels. The ESPRESSO and HARPS data have σ_K values of 1.23 and 0.82 m/s, respectively. Table 4 summarizes this information, and Figure 4 shows the four radial velocity data sets.

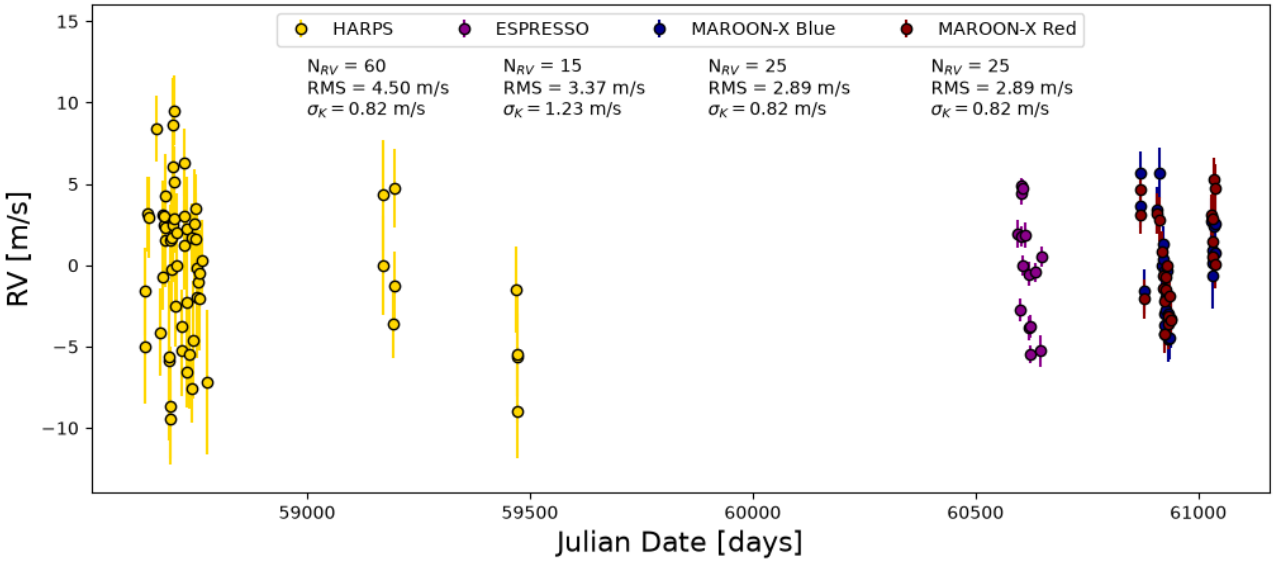


Figure 4: Radial velocities of TOI-233 extracted from HARPS (yellow), ESPRESSO (purple), and MAROON-X (red and blue). The number of data points, $N_{\rm RV}$, the RMS, and the information content, σ_K, are shown.

Parameter	MAROON-X Blue	MAROON-X Red	ESPRESSO	HARPS
$N_{\rm RV}$	25	25	15	60
$\Delta RV_{\rm mean}$ [m/s]	1.443	1.257	0.682	2.707
RMS [m/s]	2.893	2.891	3.369	4.498
σ_K [m/s]	0.818	0.817	1.230	0.821

Table 3: RVs information content parameters for each instrument.

Stellar activity

Figure 5 shows the Lomb-Scargle periodograms of the MAROON-X RV data and of Hα. Figure 6 shows the activity indicators CRX, dLW, and Hα taken from SERVAL, together with their relation to the RVs extracted with the self-template. The first two indices show no correlation with the RVs, with r very close to 0. Hα, on the other hand, shows a negative correlation, with p-values for the red and blue channels of $p_R = 0.007$ and $p_B = 0.021$, respectively. When we compute the Lomb-Scargle periodograms of these activity indices, only Hα shows a significant peak, near ~ 43 days, as seen in Figure 5. This peak coincides with the stellar rotation period ($P_{\rm rot} \sim 44$ days, HATSouth, Joel Hartman 2018), a signal that is also present in the MAROON-X data for both channels. The periodograms of these three indices, together with those of Na I, Ca I, and Ca I 2, which also show a peak in the same region, can be found in the files attached to this document. From this same figure we also recover the two planets' orbital periods in the radial velocity data, consistent with the TESS transit results, both for the individual channels and for the combined data. For this last periodogram, the purple line in Figure 5, the False Alarm Probability at the 1.0% level is FAP = 0.38.

Radial Velocity Fit

The radial velocity fit to the MAROON-X and ESPRESSO data, assuming the Keplerian model described above, a maximum-eccentricity constraint $e_{\rm cross}$, and a quasi-periodic Gaussian Process motivated by the stellar activity results already discussed, found that the most favored model, with

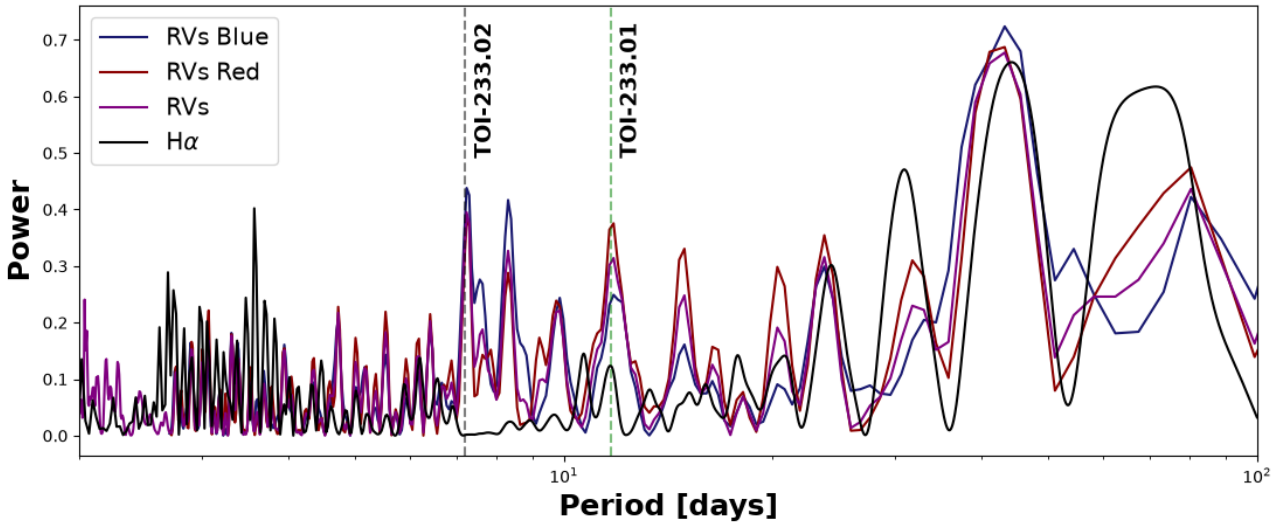


Figure 5: Periodograms of the MAROON-X RVs and of the Hα activity index. The individual channels (red and blue lines) and the combined RV data (purple line) recover both the planetary periods and the stellar rotation period. The Hα periodogram (black line) shows only one significant peak, near the stellar rotation period $P_{\rm rot} \sim 43$ days.

BIC = 306.11 and AICc = 284.98, is the one with free parameters K_1, K_2, GP, and γ, followed by the model that also includes e_1, with BIC = 311.38 and AICc = 288.56. The posteriors derived from the MCMC for the minimum masses, semi-amplitudes, and eccentricities, together with the best-fit Keplerian orbital model for the TOI-233 system, are shown in Figure 7, taken from the final RadVel report. In RadVel's notation, subscript **b** refers to planet TOI-233.01 and subscript **c** to planet TOI-233.02. For the full table of posteriors and a more detailed look at the results, we refer the reader to the RadVel report attached to this document[3]. A summary of the most important final orbital parameters is given below:

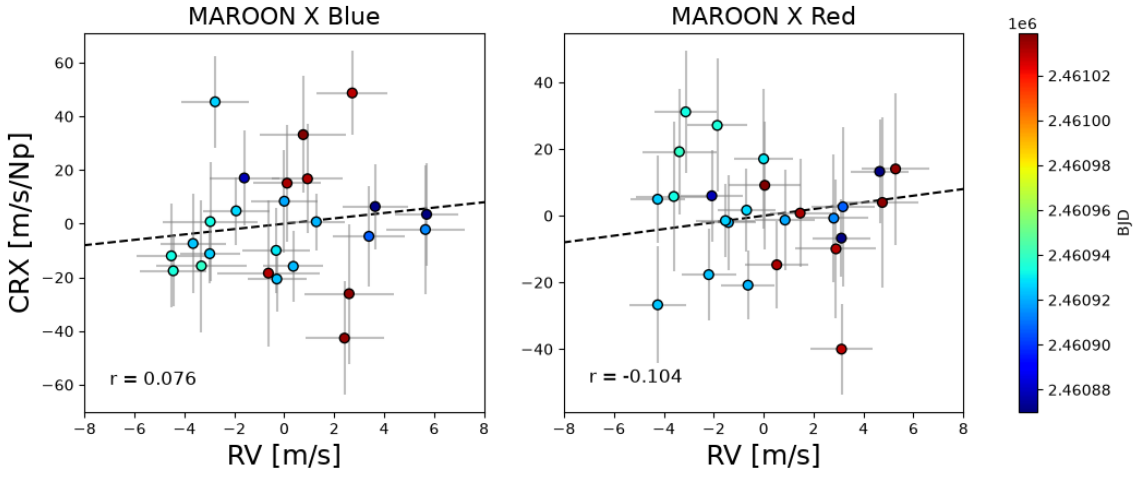
Planet	P [days]	e	ω [rad]	K [m/s]	a [AU]	$M \sin i$ [$M_\oplus$]
TOI-233.01	11.67 ± 0.0001	$0.070^{+0.057}_{-0.048}$	$-0.1^{+2.1}_{-2.0}$	$1.57^{+0.36}_{-0.39}$	$0.0723^{+0.0031}_{-0.0034}$	$2.82^{+0.72}_{-0.73}$
TOI-233.02	7.2011 ± 0.0001	$0.076^{+0.055}_{-0.052}$	$2.8^{+1.9}_{-1.6}$	1.83 ± 0.34	$0.0524^{+0.0023}_{-0.0025}$	$2.81^{+0.61}_{-0.57}$

Table 4: Posteriors and derived posteriors for the best-fit 2-planet Keplerian orbital model for TOI-233 using MAROON-X and ESPRESSO data.

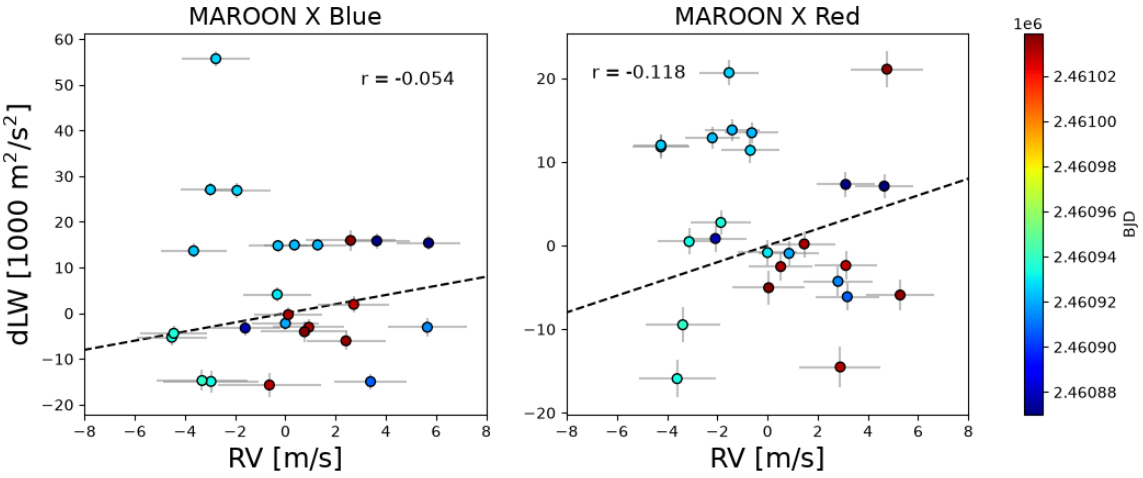
Discussion: Planetary Structure

Previous studies have found that sub-Neptunes near mean-motion resonances are puffier than those that are not near resonance (e.g., Andrien et al. 2024). The masses and radii derived for the TOI-233 system place both planets at the boundary of the rather diffuse distinction between super-Earths and sub-Neptunes. Figure 8 shows the results of the planetary composition models. The models from Aguichine et al. (2025), represented by the dashed lines, provide the best fit for both planets when varying the WMF. For TOI-233.01, the planet with the larger radius and longer orbital period, its

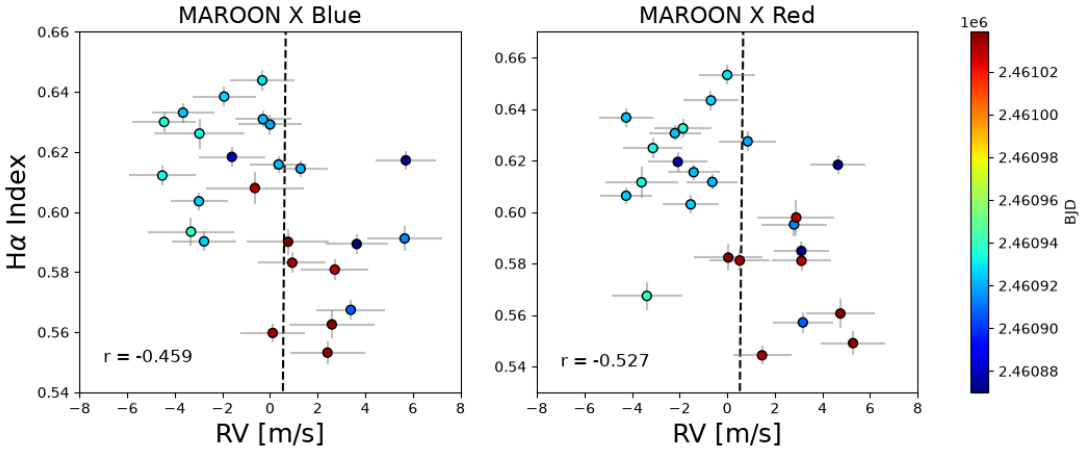
[3]See `TOI233_MX_BR_ESP_1GP_ecross_results.pdf`



(a)



(b)



(c)

Figure 6: Comparison of the MAROON-X RVs with the stellar activity indices CRX (top), dLW (middle), and Hα (bottom). The Pearson correlation coefficient is shown on each panel.

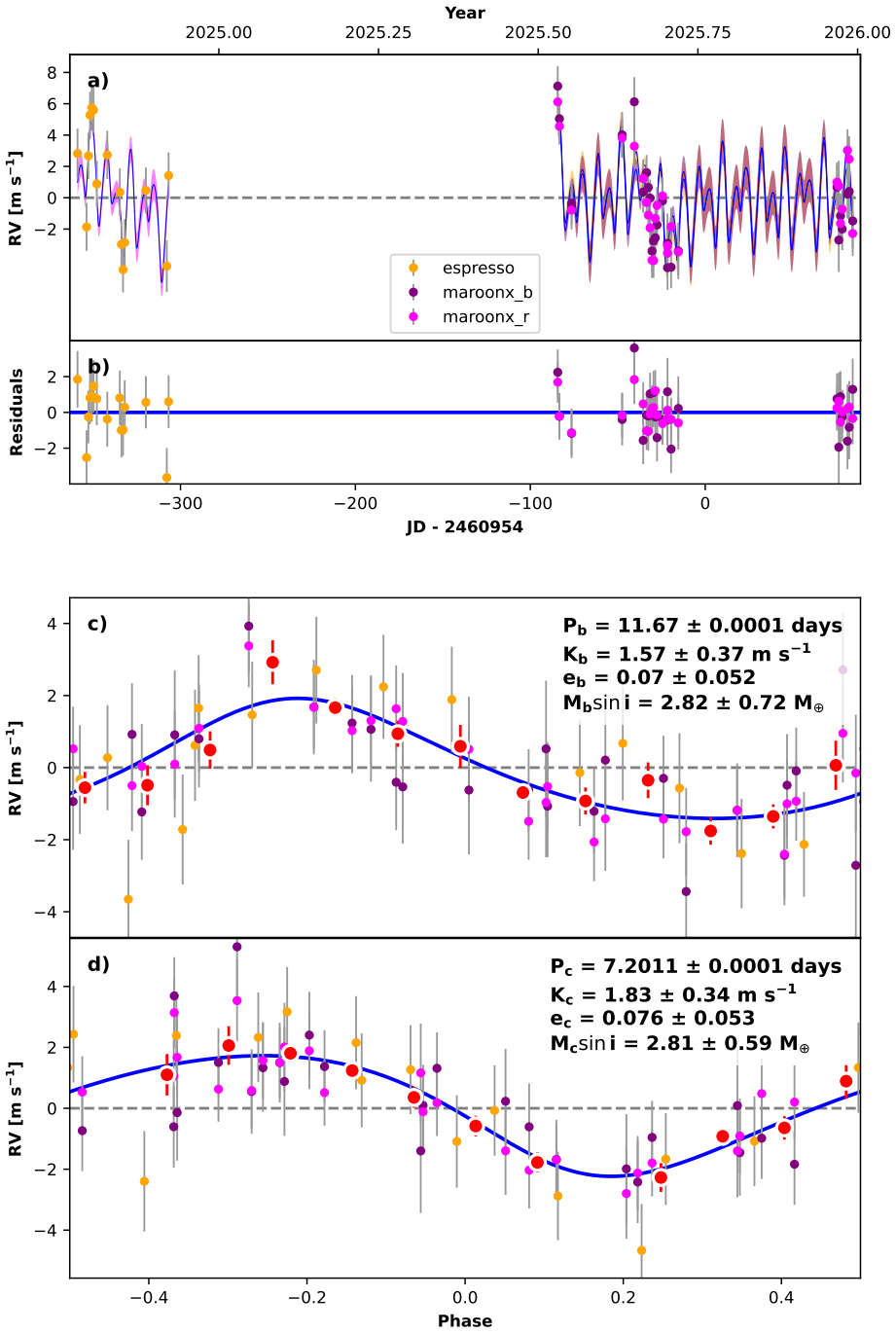


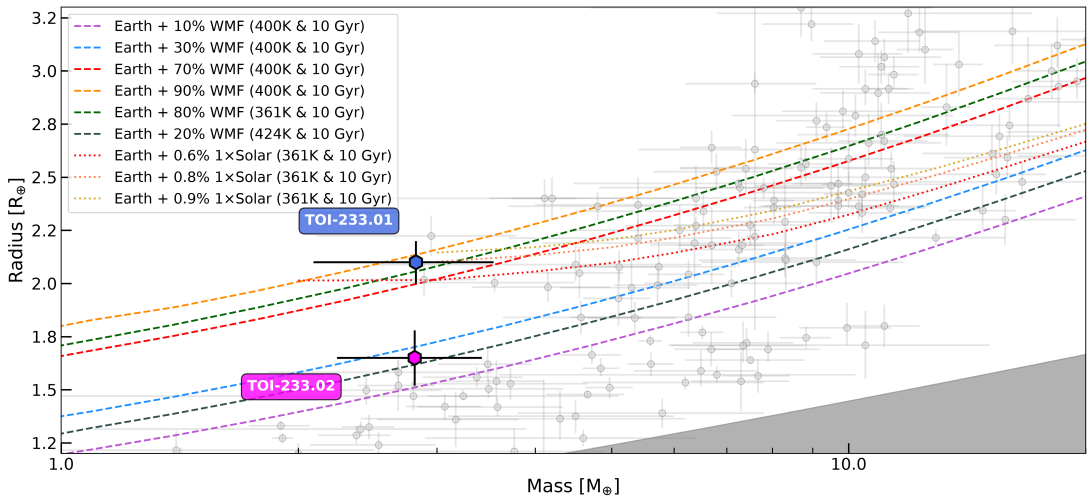
Figure 7: Best-fit 2-planet Keplerian orbital model for TOI-233. **Top: a)** The maximum likelihood model is plotted while the orbital parameters listed in Table 2* are the median values of the posterior distributions. The thin blue line is the best fit 2-planet model. We add in quadrature the RV jitter term(s) listed in Table 2* with the measurement uncertainties for all RVs. **b)** Residuals to the best fit 2-planet model. **Bottom: a)** RVs phase-folded to the ephemeris of planet b. The Keplerian orbital models for planet c has been subtracted. The small point colors and symbols are the same as in panel **a**. Red circles are the same velocities binned in 0.08 units of orbital phase. The phase-folded model for planet b is shown as the blue line. **b)** Same for planet c.

* Table 2 in this caption refers to Table 2 in the RadVel report.

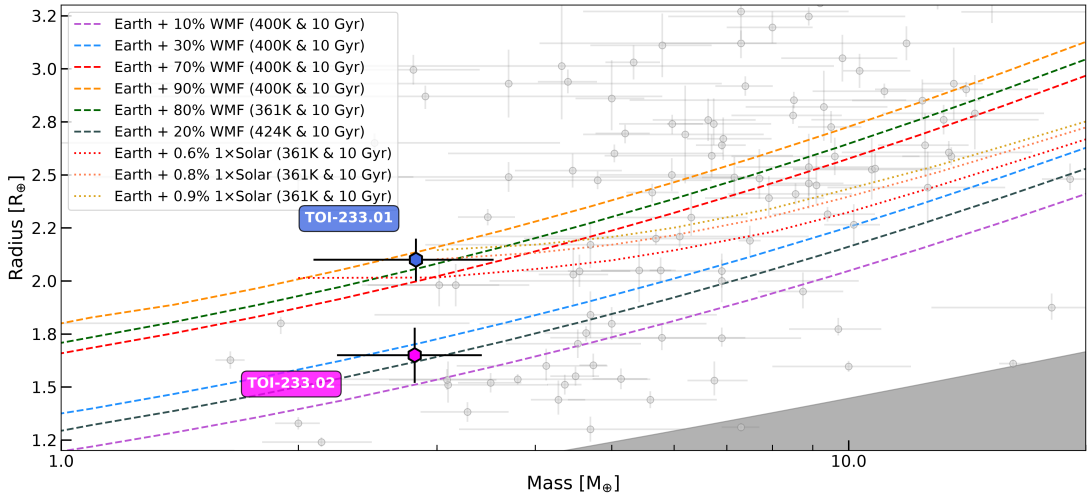
physical parameters place it in the high-WMF region, with the 70% WMF model representing its lower limit and the 90% model its upper limit, as the uncertainties in its mass and radius allow it to lie between these two cases. The model that best matches the planet uses its reported $T_{\rm eq}$ and a WMF of 80%, although the authors recommend avoiding values below 400 K due to interpolation issues.

The models from Tang et al. (2025) only marginally reproduce the physical parameters of the planet once planets that have undergone atmospheric boil-off are excluded. Models with a thinner envelope are not considered; therefore, TOI-233.02 would not correspond to a planet with a thin H_2/He envelope, but rather to a planet with a low WMF, with the 10% model representing its lower limit and the 30% model its upper limit. This, together with the well-known bimodal distribution separating super-Earths from sub-Neptunes, which can be seen in Figure 8a, where TOI-233.02 and TOI-233.01 appear to lie in the respective populations, would support the nature of their internal structures. In particular, the inner planet may have experienced greater atmospheric erosion due to its shorter distance from the host star. However, it is worth asking whether the difference in the planets' semimajor axes is sufficiently large for atmospheric erosion to affect TOI-233.01 to a lesser extent.

On the other hand, Figure 8b shows the increase in planetary radius among planets near resonance, with a clear increase in the population of planets with masses between 4–5 $M_\oplus$. Given the constraints imposed, our system lies in a sparsely populated region, making TOI-233 a new benchmark system for studying low-mass planets near mean-motion resonances.



(a)



(b)

Figure 8: TOI-233 system in the context of evolutionary models of steam worlds and gas dwarfs. The dashed lines represent the "steam worlds" models by Aguichine et al. (2025) for different values of WMF and temperatures. The dotted lines represent the "gas dwarf" models by Tang et al. (2025). The gray dots are confirmed exoplanets that are not in resonance (**a**) and those that are in resonance (**b**) in multiplanetary systems with measured periods and masses smaller than that of Jupiter.

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